

Graph Reconstruction from the Observation of Diffused Signals

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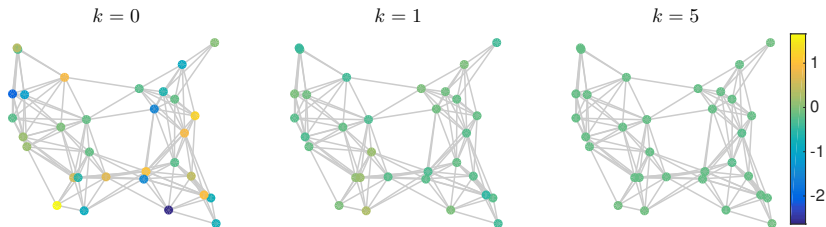
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Recovering Graph Structure

Given indirect observations of signals diffused over a graph, estimate the graph structure.



Related Work

- Gaussian graphical models and covariance selection
 - Covariance selection: Dempster (1972)
 - Sparsity:
 - Meinshausen and Bühlman (2006)
 - Friedman, Hastie, and Tibshirani (2008)
 - Computationally efficient algorithms:
 - Bannerjee, El Ghaoui, d'Aspremont (2008)
 - Hsieh, Sustik, Dhillon, & Ravikumar (2012)
 - Bayesian: Costard (2014)
- From a graph signal processing perspective
 - Dong, Thanou, Frossard, & Vandergheynst (2014)
 - Zhang, Florêncio, & Chou (2015)

Laplacian and Diffusion Matrices

- Let $G = (V, E)$ be the **unknown** undirected graph
 - with $|V| = n$ vertices,
 - adjacency matrix A ,
 - and (diagonal) degree matrix D
- The **normalized Laplacian** matrix is

$$L = D^{-1/2}(D - A)D^{-1/2}$$

- The **diffusion matrix** is

$$\begin{aligned} T &= I - L \\ &= D^{-1/2}AD^{-1/2} \end{aligned}$$

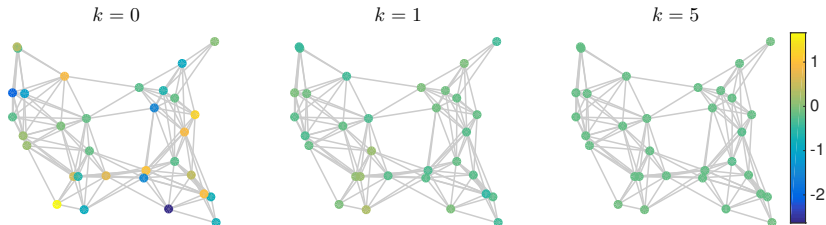
Diffusion Matrix and Diffused Signals

Why **diffusion matrix**? $T = D^{-1/2}AD^{-1/2}$

Consider a signal $x(t) \in \mathbb{R}^V$ diffusing over G

Discretization of the heat equation $\frac{\partial x}{\partial t} = -Lx$ gives

$$x_{k+1} - x_k = -Lx_k \implies x_k = Tx_{k-1} = T^k x_0$$



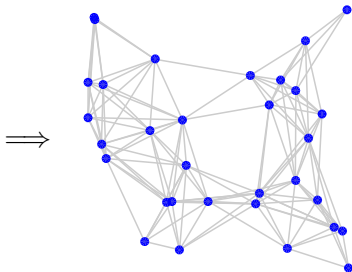
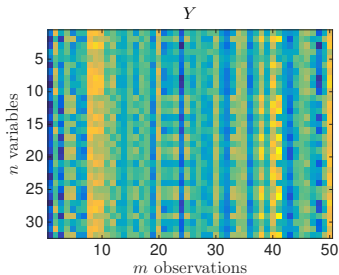
Problem Formulation

Let $x_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I)$ for $i = 1, \dots, m$

Observe: y_i , $i = 1, \dots, m$, where $y_i = T^k x_i$
(Assume k is known)

Combine observations into data matrix $Y = [y_1, \dots, y_m]$

Goal: Estimate T (as a proxy for G)



Covariance Matrix Estimation

If $x_i \sim \mathcal{N}(0, I)$ then $y_i = T^k x_i \sim \mathcal{N}(0, T^{2k})$

Sample covariance matrix $S_m = \frac{1}{m} Y Y^\top \rightarrow T^{2k}$ as $m \rightarrow \infty$

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Note: T^{2k} may not be sparse, even if T is

- Depends on the diameter of G relative to $2k$
- If sparse, use better/appropriate sparse covariance estimators

A Simpler(?) Problem

Given T^{2k} and k , recover T

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When is this possible?

G is undirected $\Rightarrow T$ is symmetric

- $T = U\Lambda U^\top$
- $T^{2k} = U\Lambda^{2k}U^\top$
- Eigenvalues of T satisfy $-1 \leq \lambda_i(T) \leq 1$
($T = I - L$, and $0 \leq \lambda_i(L) \leq 2$)
- **Problem:** No information about eigenvalue signs from T^{2k}

Recovering Eigenvalue Signs

Let $T^{2k} = U\Xi U^\top$, with eigenvalues $\xi_1, \xi_2, \dots, \xi_n$

Then $T = U\Lambda U^\top$ where $\lambda_i = \pm \xi_i^{-1/2k}$

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Properties of diffusion matrices $T = D^{-1/2}AD^{-1/2}$

- $T_{i,i} = 0$ since A is simple
- $T_{i,j} \geq 0$ for $i \neq j$

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- $T_{i,i} = 0$ since A is simple
- $T_{i,j} \geq 0$ for $i \neq j$
- $\max_i \lambda_i(T) = 1$
- If G is connected then exactly one $\lambda_i(T) = 1$
- If G is not bipartite then $\lambda_i(T) > -1$ for all i

Relax to a Linear Program

Given U and $1 = \xi_1, \xi_2, \dots, \xi_n$,

Find $\lambda \in \mathbb{R}^n$ that solves

$$\begin{aligned} & \text{minimize} && \max_{i=1,\dots,n} \left| e_i^\top U \text{diag}(\lambda) U^\top e_i \right| \\ & \text{subject to} && \lambda_1 = 1 \\ & && -\xi_i^{1/2k} \leq \lambda_i \leq \xi_i^{1/2k}, \quad i = 2, \dots, n \\ & && e_i^\top U \text{diag}(\lambda) U^\top e_j \geq 0, \quad i \neq j \end{aligned}$$

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Note: All constraints are linear in λ

Many off-diagonal constraints are redundant

When is there a Unique Solution?

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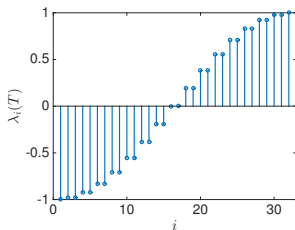
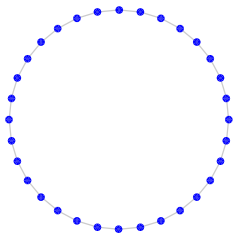
Conjecture: If the eigenvalue magnitudes $|\lambda_1|, |\lambda_2|, \dots, |\lambda_n|$ are distinct, then the linear program has a unique solution

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An example where things break: the **ring graph** (with n even)



Recovering Graph Structure from Diffused Signals

Ongoing work and questions:

- Use sparsity of $\hat{T}$ to resolve non-uniqueness
- Integrating estimation of T^{2k} and recovery of eigenvalue signs
- What if k is unknown?
- What if G is not connected?

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